HW0303

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4				
2	2				
1	3				
−1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum(x_i-\bar{x}) =$	$\sum(x_i-\bar{x})^2 =$	$\sum(y_i-\bar{y}) =$	$\sum(x_i-\bar{x})(y_i-\bar{y}) =$

a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

c. Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$. Using these numerical values, show that $\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

d. Use the least squares estimates from part (b) to compute the fitted values of y, and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y, $s_y^2 = \sum_{i=1}^{N}(y_i-\bar{y})^2/(N-1)$, the sample variance of x, $s_x^2 = \sum_{i=1}^{N}(x_i-\bar{x})^2/(N-1)$, the sample covariance between x and y, $s_{xy} = \sum_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})/(N-1)$, the sample correlation between x and y, $r_{xy} = s_{xy}/(s_x s_y)$ and the coefficient of variation of x, $CV_x = 100(s_x/\bar{x})$. What is the median, 50th percentile, of x?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4				
2	2				
1	3				
−1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$	$\sum \hat{e}_i =$	$\sum \hat{e}_i^2 =$	$\sum x_i\hat{e}_i =$

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.

f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i/N$.

i. Compute $\hat{\sigma}^2$.

j. Compute $\widehat{\text{var}}(b_2|\mathbf{x})$ and $\text{se}(b_2)$.

a.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
−1	1	−2	4	−1	2
0	0	−1	1	−2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum(x_i-\bar{x}) = 0$	$\sum(x_i-\bar{x})^2 = 10$	$\sum(y_i-\bar{y}) = 0$	$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8$

$\Rightarrow \bar{x} = \frac{5}{5} = 1$ $\quad \Rightarrow \bar{y} = \frac{10}{5} = 2$

b.

$$b_2 = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \frac{8}{10} = 0.8$$

$\Rightarrow$ slope

$$b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8\times 1 = 1.2.$$

$\Rightarrow$ intercept

$$y_i = 1.2 + 0.8x_i$$

c.

$$\sum_{i=1}^{5} x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$$

$$\sum_{i=1}^{5} x_i y_i = 3\times4 + 2\times2 + 1\times3 + (-1)\times1 + 0\times0 = 18$$

$$\sum(x-\bar{x})^2 = \sum x^2 - N\bar{x}^2$$

$$\Rightarrow 10 = 15 - 5\times1$$

$$\sum(x-\bar{x})(y-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$$

$$\Rightarrow 8 = 18 - 5\times1\times2$$

d. $\hat{y}_i = 1.2 + 0.8x \Rightarrow$

$\hat{e}_i = y - \hat{y}_i$

x	y	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	0.6
0	0	1.2	-1.2	1.44	0
$\Sigma x_i = 5$	$\Sigma y_i = 10$	$\Sigma \hat{y}_i = 10$	$\Sigma \hat{e}_i = 0$	$\Sigma \hat{e}_i^2 = 3.6$	$\Sigma x_i \hat{e}_i = 0$

$$S_y^2 = \sum_{i=1}^{N} \frac{(y_i - \bar{y})^2}{N-1} = \frac{2^2 + 0^2 + 1^2 + (-1)^2 + (-2)^2}{5-1} = \frac{10}{4} = 2.5$$

$$S_x^2 = \sum_{i=1}^{N} \frac{(x_i - \bar{x})^2}{N-1} = \frac{4+1+0+4+1}{5-1} = \frac{10}{4} = 2.5$$

$$S_{xy} = \sum_{i=1}^{N} \frac{(x_i - \bar{x})(y_i - \bar{y})}{N-1} = \frac{4+0+0+2+2}{5-1} = \frac{8}{4} = 2$$

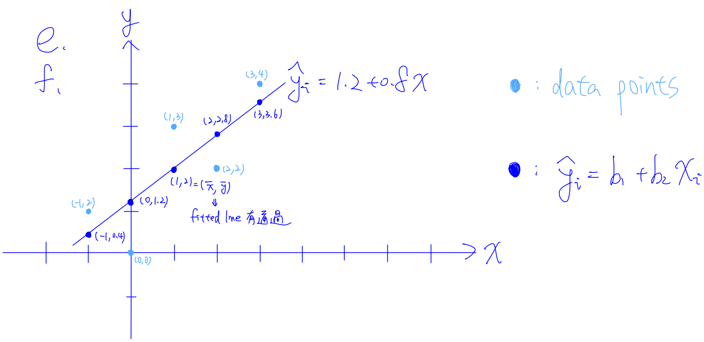
$$r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{\sqrt{2.5}\sqrt{2.5}} = 0.8$$

$$CV_x = 100 \times \frac{S_x}{\bar{x}} = 100 \times \frac{\sqrt{2.5}}{1} = 158.1139$$

$x = \{-1, 0, 1, 2, 3\} \Rightarrow$ median of $x = 1$

e.

f.



g. $2 = 1.2 + 0.8x_1$

h. $\bar{\hat{y}} = \frac{\Sigma \hat{y}_i}{N} = \frac{10}{5} = 2$

i. $\hat{\sigma}^2 = \frac{\Sigma \hat{e}_i^2}{N} = \frac{3.6}{5} = 0.72$

j. $\widehat{var}(b_2 | x) = \frac{\hat{\sigma}^2}{\Sigma(x_i - \bar{x})^2} = \frac{0.72}{10} = 0.072$

$se(b_2) = \sqrt{\widehat{var}(b_2|x)} = \sqrt{0.072} = 0.2683$

2.22 A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

a. Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?

b. Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

c. Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?

d. Repeat part (c) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

a.

```
> library(POE5Rdata)
> data("star5_small")
> ?star5_small
>
> # subset data
> data_a <- subset(star5_small, small == 1 | regular== 1)
>
> # regression
> model_a <- lm(totalscore ~ small, data = data_a)
>
> summary(model_a)

Call:
lm(formula = totalscore ~ small, data = data_a)

Residuals:
    Min      1Q  Median      3Q     Max
-189.44  -50.44   -9.44   43.06  324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442      3.675 249.396   <2e-16 ***
small         12.175      5.369   2.268   0.0236 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608,	Adjusted R-squared:  0.005323
F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

$TOTALSCORE_i = 916.442 + 12.175 SMALL_i$

估計出的截距项是916.442

⇒當SMALL=0，即是regular class時，學生的平均TOTALSCORE

估計出的SMALL係數是12.175

⇒在small class的學生TOTALSCORE比在regular class的高12.175分.

p=0.0236 ⇒ 此係數在5%的顯著水準下有統計顯著性.

⇒迴歸顯示較小的班級在學生的學習成果有正面影響.

b.

```
> # reading score
> model_b1 <- lm(readscore ~ small, data = data_a)
> summary(model_b1)

Call:
lm(formula = readscore ~ small, data = data_a)

Residuals:
    Min      1Q  Median      3Q     Max
-74.665 -21.590  -2.665  16.410 194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665      1.488 290.760  < 2e-16 ***
small          6.924      2.174   3.185  0.00151 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295,	Adjusted R-squared:  0.01167
F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

$READSCORE_i = 432.665 + 6.924 SMALL_i$

估計出的截距项是432.665 ⇒ regular class學生的平均READSCORE

估計出的SMALL係數是6.924 ⇒ small class學生的READSCORE平均高 6.924分

p=0.00151 ⇒ small class對閱讀成績的影響在1%的顯著水準下有統計顯著性.

```
> # math score
> model_b2 <- lm(mathscore ~ small, data = data_a)
> summary(model_b2)

Call:
lm(formula = mathscore ~ small, data = data_a)

Residuals:
     Min       1Q   Median       3Q      Max
-144.777  -35.028   -5.028   29.223  142.223

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777      2.449 197.545   <2e-16 ***
small          5.251      3.578   1.467    0.143
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778,	Adjusted R-squared:  0.001488
F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

$MATHSCORE_i = 483.777 + 5.251 SMALL_i$

估計出的截距项是483.777 ⇒ regular class學生的平均MATHSCORE

估計出的SMALL係數是5.251 ⇒ small class學生的MATH SCORE平均高5.251 分

p=0.143 ⇒ small class對數學成績不具統計顯著性.

⇒和(a)有差異.

c.

```
> data_c <- subset(star5_small, regular == 1 | aide == 1)
> model_c <- lm(totalscore ~ aide, data = data_c)
> summary(model_c)

Call:
lm(formula = totalscore ~ aide, data = data_c)

Residuals:
     Min       1Q   Median       3Q      Max 
-191.748  -50.442   -5.442   40.558  312.558 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.559 257.529   <2e-16 ***
aide           4.306      4.994   0.862    0.389    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898,	Adjusted R-squared:  -0.0003068 
F-statistic: 0.7436 on 1 and 835 DF,  p-value: 0.3888
```

$TOTALSCORE_i = 916.442 + 4.306\,AIDE$

估計出的截距項是916.442

⇒當 regular class 沒有 teacher aide 時，學生的平均 TOTALSCORE

估計出的AIDE係數是4.306

⇒在有 teacher aide 的 regular class 學生 TOTALSCORE
比在沒有 teacher aide 的 regular class 學生成績高12.175分.

p=0.389 ⇒ 此係數不具統計顯著性.

⇒無法顯示 teacher aide 對學生的學習成果有正面影響.

d.

```
> # reading score with aide
> model_d1 <- lm(readscore ~ aide, data = data_c)
> summary(model_d1)

Call:
lm(formula = readscore ~ aide, data = data_c)

Residuals:
    Min      1Q  Median      3Q     Max 
-74.665 -21.536  -2.536  15.464 194.335 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.460 296.341   <2e-16 ***
aide           2.871      2.049   1.401    0.161    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared:  0.002347,	Adjusted R-squared:  0.001152 
F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615
```

$TOTALSCORE_i = 432.665 + 2.871\,AIDE$

估計出的截距項是432.665 ⇒ 沒有 teacher aide 的 regular class 學生的平均 READSCORE

估計出的SMALL係數是2.871 ⇒ 有 teacher aide 的班級學生的READSCORE平均高2.871分

p=0.161 ⇒ teacher aide 對閱讀成績的影響不具統計顯著性.

```
> # math score with aide
> model_d2 <- lm(mathscore ~ aide, data = data_c)
> summary(model_d2)

Call:
lm(formula = mathscore ~ aide, data = data_c)

Residuals:
     Min       1Q   Median       3Q      Max 
-146.212  -31.212   -5.777   29.223  142.223 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.355 205.464   <2e-16 ***
aide           1.435      3.304   0.434    0.664    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared:  0.0002258,	Adjusted R-squared:  -0.0009715 
F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

$TOTALSCORE_i = 483.777 + 1.435\,AIDE$

估計出的截距項是483.777 ⇒ 沒有 teacher aide 的 regular class 學生的平均 MATHSCORE

估計出的SMALL係數是 1.435 ⇒ 有 teacher aide 的班級學生的MATHSCORE平均高1.435 分

p=0.664 ⇒ teacher aide 對數學成績的影響不具統計顯著性.

⇒兩種迴歸結果和(c)一致.

2.25 Consumer expenditure data from 2013 are contained in the file *cex5_small*. [Note: *cex5* is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between \$1000 per month to \$20,000 per month. *FOODAWAY* is past quarter's food away from home expenditure per month per person, in dollars, and *INCOME* is household monthly income during past year, in \$100 units.

a. Construct a histogram of *FOODAWAY* and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?

b. What are the mean and median values of *FOODAWAY* for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?

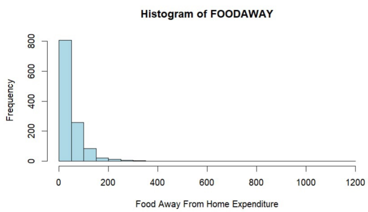
c. Construct a histogram of ln(*FOODAWAY*) and its summary statistics. Explain why *FOODAWAY* and ln(*FOODAWAY*) have different numbers of observations.

d. Estimate the linear regression $\ln(FOODAWAY) = \beta_1 + \beta_2 INCOME + e$. Interpret the estimated slope.

e. Plot ln(*FOODAWAY*) against *INCOME*, and include the fitted line from part (d).

f. Calculate the least squares residuals from the estimation in part (d). Plot them vs. *INCOME*. Do you find any unusual patterns, or do they seem completely random?

a.

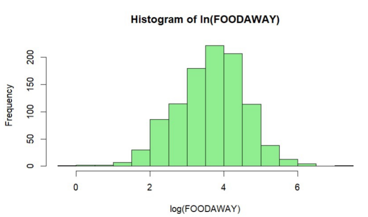



```
> # Histogram
> hist(cex5_small$foodaway,
+      main = "Histogram of FOODAWAY",
+      xlab = "Food Away From Home Expenditure",
+      col = "lightblue",
+      breaks = 20)
>
> # summary statistics
> summary(cex5_small$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   12.04   32.55   49.27   67.50 1179.00
>
> # mean
> mean(cex5_small$foodaway, na.rm = TRUE)
[1] 49.27085
>
> # median
> median(cex5_small$foodaway, na.rm = TRUE)
[1] 32.555
>
> # 25th and 75th percentiles
> quantile(cex5_small$foodaway, probs = c(0.25,0.75), na.rm = TRUE)
    25%     75%
12.0400 67.5025
```

b.

```
> # advanced degree households
> mean(cex5_small$foodaway[cex5_small$advanced == 1], na.rm=TRUE)
[1] 73.15494
> median(cex5_small$foodaway[cex5_small$advanced == 1], na.rm=TRUE)
[1] 48.15
>
> # college degree households
> mean(cex5_small$foodaway[cex5_small$college == 1], na.rm=TRUE)
[1] 48.59718
> median(cex5_small$foodaway[cex5_small$college == 1], na.rm=TRUE)
[1] 36.11
>
> # no college or advanced degree
> mean(cex5_small$foodaway[cex5_small$advanced == 0 & cex5_small$college == 0], na.rm=TRUE)
[1] 39.01017
> median(cex5_small$foodaway[cex5_small$advanced == 0 & cex5_small$college == 0], na.rm=TRUE)
[1] 26.02
```

c.




```
> # 建立 log variable
> cex5_small$lnfoodaway <- log(cex5_small$foodaway)
>
> # histogram
> hist(cex5_small$lnfoodaway,
+      main="Histogram of ln(FOODAWAY)",
+      xlab="log(FOODAWAY)",
+      col="lightgreen",
+      breaks=20)
>
> # summary statistics
> summary(cex5_small$lnfoodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   -Inf   2.488   3.483    -Inf   4.212   7.072
```

因為log(N)在N≤0時沒有值,被排除在樣本外,造成兩組的樣本數不同.

Made with Goodnotes

d.

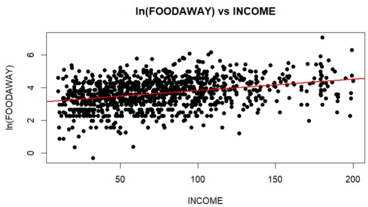
```
> model <- lm(log(foodaway) ~ income, data = cex5_small, subset = foodaway > 0)
> summary(model)

Call:
lm(formula = log(foodaway) ~ income, data = cex5_small, subset = foodaway >
    0)

Residuals:
    Min      1Q  Median      3Q     Max
-3.6547 -0.5777  0.0530  0.5937  2.7000

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826,	Adjusted R-squared:  0.09738
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

$\ln(FOODAWAY) = 3.1293 + 0.0069017\, INCOME$.

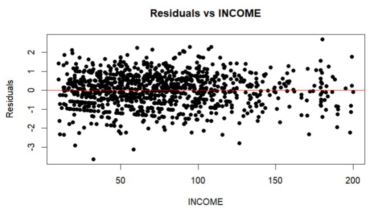
slope = 0.0069017

⇒ INCOME 增加100時, FOODAWAY 增加 0.69017%

e.



f.



residuals 趨勢呈現水平線

表示 OLS 模型有效.